Gauge–Gravity Spectral Synthesis: Joint Einstein–Yang–Mills Equations, Born–Infeld Completion, and the Hierarchy Ratio from Projective Spectral Entropy

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Abstract

The companion papers Gravity 3.0 [7] and Q12 [11] derived, respectively, the Einstein tensor as the horizontal a_2 response and the Yang–Mills equations as the vertical a_4 response of the projective spectral entropy functional $S_\Pi[g, A] = \frac{1}{2}\log\det' \mathcal{A}_{g,A}$. The present paper carries out the gauge–gravity synthesis and establishes its central conceptual consequence: *gravity and gauge dynamics are not unified by enlarging the symmetry group or adding fields, but are stratified by the Seeley–DeWitt level at which the same admissible operator responds to variation.* The conventional question "what symmetry unifies gravity and gauge?" is thereby replaced by "at what spectral order does the projection respond?"

We solve the *joint* variational problem $\delta_{g,A} S_\Pi = 0$ and derive the coupled Einstein–Yang–Mills system: the Yang–Mills stress tensor sources gravity through the metric dependence of the a_4 coefficient, with coupling constant $8\pi G_N = c_{YM}/c_{EH}$ fixed by the Seeley–DeWitt expansion alone. We compute the a_6 coefficient and identify the leading gauge–gravity cross-coupling $R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$. The Eddington–Born–Infeld completion of the joint functional is constructed from the same Born–Infeld saturation bound that governs the gravitational sector [3]. The gauge–gravity hierarchy is a structural output of the spectral stratification: gravity is induced at a_2 (quadratic UV divergence) and gauge at a_4 (logarithmic UV divergence), from the same cutoff $\ell_{\rm sp}$, giving:

$$G_N g_{YM}^2 \;\sim\; \frac{\ell_{\rm sp}^2}{\dim V} \cdot \left[\log\frac{\Lambda}{\mu}\right]^{-1} \;\ll\; 1,$$

without fine-tuning. The spectral stratification $a_2 \to$ gravity, $a_4 \to$ gauge, $a_6 \to$ mixed coupling predicts that fermionic structure should appear at a dedicated spectral level carrying a Dirac-type admissible operator, which is the central open question for the continuation of the programme. The gauge group G_Π is inherited from the spectral admissibility sub-programme [6, 10]; the SU(3) sector retains the conditional status of hypothesis [H-color].

Keywords: gauge–gravity synthesis, Einstein–Yang–Mills equations, Seeley–DeWitt expansion, spectral entropy functional, Eddington–Born–Infeld completion, gauge–gravity hierarchy, non-injective projection, admissible bundle.

Contents

1 Introduction

1.1 Context

The Cosmochrony framework derives emergent physics from a generally non-injective projection $\Pi\colon \chi \to \mathcal{M}$ of a static relational substrate χ. Within this framework, the Q-series of papers establishes the emergence of quantum structure, spacetime geometry, and gauge dynamics from the admissibility structure of Π [2, 5, 8]. The spectral entropy functional

$$S_\Pi[g] = \tfrac{1}{2} \log {\det}' \, \mathcal{A}_g,$$

where $\mathcal{A}_g = -(\nabla^g)^2$ is the admissible Laplacian, encodes the effective geometric dynamics of the projection. Its metric variation produces the Einstein tensor as the infrared-dominant local response via the a_2 Seeley–DeWitt coefficient [7].

The Q12 paper [11] extended the functional to the gauge sector by coupling $\mathcal{A}_g$ to an admissible connection A on the associated vector bundle:

$$S_\Pi[g, A] = \tfrac{1}{2} \log {\det}' \, \mathcal{A}_{g,A}, \qquad \mathcal{A}_{g,A} = -(\nabla^A)^2 + E.$$

The vertical variation $\delta_A S_\Pi = 0$ then yields the Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ through the a_4 coefficient. The two results place gravity and Yang–Mills dynamics at different levels of the Seeley–DeWitt hierarchy of the *same* functional, separated by geometric direction (horizontal versus vertical variation) and by spectral order.

1.2 Main results

The present paper delivers the four items deferred from Q12 [11]:

1. **Joint variational problem.** We solve $\delta_{g,A} S_\Pi = 0$ as a coupled system and show that the Yang–Mills stress tensor $T^{\mathrm{YM}}_{\mu\nu}$ enters the metric equation as the unique source term produced by the a_4 coefficient under metric variation (Theorem 3.2).

2. **Gauge–gravity coupling at order a_6.** We compute the leading cross-term in the a_6 Seeley–DeWitt coefficient, which contains $R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$ and encodes the gravitational backreaction of the gauge field at higher spectral order (Theorem 4.1).

3. **Eddington–Born–Infeld completion.** We construct the EBI extension of $S_\Pi[g, A]$ from the same saturation bound c_χ that underlies the gravitational Born–Infeld completion of [3] (Theorem 5.3).

4. **Quantitative hierarchy ratio.** We derive a precise structural bound on $G_N g^2_{YM}$ and show that it is structurally suppressed in both ℓ^2_{sp} and $[\log(\Lambda/\mu)]^{-1}$, without fine-tuning (Proposition 6.1).

1.3 Organisation

Section 2 states the joint functional and recalls the horizontal–vertical decoupling structure. Section 3 derives the Einstein–Yang–Mills system. Section 4 computes the a_6 cross-coupling. Section 5 constructs the Eddington–Born–Infeld completion. Section 6 derives the hierarchy ratio quantitatively. Section 7 discusses the epistemic status and conditionalities. Section 8 concludes.

2 The joint spectral entropy functional

2.1 The admissible principal bundle

Following Q12 [11], let (M,g) be a compact Riemannian four-manifold and let $P \to M$ be a principal G_Π-bundle, where G_Π is the admissible gauge group inherited from the spectral admissibility sub-programme [6]. Let $E = P \times_\rho V$ be the associated vector bundle for a representation $\rho\colon G_\Pi \to \mathrm{GL}(V)$. An admissible connection $A \in \mathcal{A}(P)$ defines a covariant derivative ∇^A on E and a curvature two-form $F_A = dA + A \wedge A \in \Omega^2(M, \operatorname{ad} P)$.

2.2 The Laplace-type operator and its decomposition

Define the Laplace-type operator on sections of E:

$$\mathcal{A}_{g,A} = -(\nabla^A)^2 + E,$$

where $E \in \Gamma(\operatorname{End} E)$ is the endomorphism determined by the admissibility structure. By the horizontal–vertical decoupling lemma of Q12 (Lemma 1 of [11]), the operator splits as:

$$\mathcal{A}_{g,A} = \mathcal{A}_g^{\mathrm{hor}} \otimes \mathrm{Id}_V \;+\; \mathrm{Id}_{\mathcal{H}} \otimes \mathcal{A}_A^{\mathrm{vert}} \;+\; \mathcal{A}^{\mathrm{mix}},$$

where $\mathcal{A}^{\mathrm{mix}}$ is suppressed at the spectral scale ℓ_{sp} by minimal coupling. The spectral entropy functional is then:

$$S_\Pi[g,A] = \tfrac{1}{2}\mathrm{Tr}\log \mathcal{A}_{g,A} \;=\; \tfrac{1}{2}\log{\det}' \,\mathcal{A}_{g,A}.$$

This is the common generating functional for both the gravitational and gauge sectors.

2.3 Seeley–DeWitt expansion of the joint functional

The heat-kernel expansion of $\mathcal{A}_{g,A}$ gives [13, 12]:

$$\mathrm{Tr}\, e^{-t\mathcal{A}_{g,A}} \;\sim\; \sum_{k=0}^{\infty} a_{2k}[\mathcal{A}_{g,A}]\, t^{(2k-4)/2} \qquad (t \to 0^+), \tag{1}$$

where the Seeley–DeWitt coefficients a_{2k} are local integrals of g, A, and their covariant derivatives. After renormalization of the UV-divergent terms, the functional decomposes as:

$$S_\Pi[g,A] = c_{(0)}\Lambda^4 a_0 \;+\; c_{(2)}\Lambda^2 a_2[g] \;+\; c_{(4)}\log\frac{\Lambda}{\mu}\, a_4[g,A] \;+\; S_\Pi^{\mathrm{fin}}[g,A] \;+\; O\!\left(\frac{1}{\Lambda^2}\right), \tag{2}$$

where S_Π^{fin} is the finite part at the renormalization scale μ. The cosmological-constant sector a_0 does not contribute to the equations of motion and is set aside. The two dynamically active sectors are:

$$a_2[g] = \frac{1}{16\pi^2}\int_M \frac{R}{6}\,\mathrm{tr}(\mathrm{Id}_V)\,\sqrt{g}\,d^4x, \tag{3}$$

$$a_4[g,A] = \frac{1}{16\pi^2}\int_M \Big[\alpha_1 R^2 + \alpha_2 R_{\mu\nu}R^{\mu\nu} + \alpha_3 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} + \tfrac{1}{12}\mathrm{Tr}(\Omega_A^2)\Big]\sqrt{g}\,d^4x, \tag{4}$$

where $\Omega_A^{\mu\nu} = F^a_{\mu\nu}T^a$ is the curvature two-form in the representation ρ and $\alpha_1, \alpha_2, \alpha_3$ are numerical coefficients depending on the spectrum of $\mathcal{A}_g$ [13]. The Yang–Mills term $\frac{1}{12}\mathrm{Tr}(\Omega_A^2)$ is the unique a_4 contribution that is first-order in the curvature two-form.

3 Joint Einstein–Yang–Mills equations of motion

3.1 Metric variation of the joint functional

The metric variation of $S_\Pi[g, A]$ receives contributions from all Seeley–DeWitt coefficients that depend on g. At leading infrared order, the dominant term is:

$$\frac{\delta S_\Pi}{\delta g^{\mu\nu}} = c_{EH}\, G_{\mu\nu}\sqrt{g} + c_{YM}\,\frac{\delta}{\delta g^{\mu\nu}}\int_M \mathrm{Tr}(\Omega_A^2)\sqrt{g}\, d^4x + O(R^3,\, R\cdot F^4), \tag{5}$$

where $c_{EH} = c_{(2)}\Lambda^2/(96\pi^2)$ and $c_{YM} = c_{(4)}\log(\Lambda/\mu)/(192\pi^2)$ are the induced coupling constants established in [7] and [11] respectively.

Lemma 3.1 (Metric variation of the Yang–Mills term)**.** *Let* $\mathcal{L}_{YM} = -\frac{1}{4}\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\sqrt{g}$. *Then:*

$$\frac{1}{\sqrt{g}}\frac{\delta}{\delta g^{\mu\nu}}\int_M \mathrm{Tr}(\Omega_A^2)\sqrt{g}\, d^4x = -4\,\mathrm{Tr}(F_{\mu\rho}F_\nu{}^\rho) + g_{\mu\nu}\,\mathrm{Tr}(F_{\rho\sigma}F^{\rho\sigma}) \equiv T^{YM}_{\mu\nu}.$$

Proof. Standard variation of the Yang–Mills kinetic term with respect to the metric, using $\partial_\alpha\sqrt{g} = \frac{1}{2}\sqrt{g}\, g^{\mu\nu}\partial_\alpha g_{\mu\nu}$ and the fact that $F_{\mu\nu}$ is metric-independent. The result is the standard Yang–Mills energy-momentum tensor. □

3.2 Gauge-field variation

From Theorem 1 of Q12 [11], the variation of $S_\Pi[g, A]$ with respect to the admissible connection A yields:

$$\frac{\delta S_\Pi}{\delta A^a_\nu} = c_{YM}\, D_\mu F^{a\mu\nu}\sqrt{g} + O(F^3,\, R\cdot F),$$

where $D_\mu = \partial_\mu + [A_\mu, \cdot\,]$ is the gauge-covariant derivative. The Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ follow at leading order.

3.3 The joint system

Theorem 3.2 (Joint Einstein–Yang–Mills equations)**.** *Let* $S_\Pi[g, A] = \frac{1}{2}\log\det{}'\,\mathcal{A}_{g,A}$ *be the joint projective spectral entropy functional. The joint variational problem* $\delta_{g,A}S_\Pi = 0$*, at leading order in the Seeley–DeWitt expansion, is equivalent to:*

$$G_{\mu\nu} = 8\pi G_N\, T^{YM}_{\mu\nu}, \tag{6}$$

$$D_\mu F^{a\mu\nu} = 0, \tag{7}$$

where the coupling constant is defined by:

$$8\pi G_N = \frac{c_{YM}}{c_{EH}}.$$

Proof. Setting $\delta_g S_\Pi = 0$ using (5) and Lemma 3.1:

$$c_{EH}\, G_{\mu\nu} + c_{YM}\, T^{YM}_{\mu\nu} = 0.$$

The standard form ([6]) follows by identifying $8\pi G_N = c_{YM}/c_{EH}$. Consistency with the explicit induced values established in [7, 11]:

$$G_N^{-1} \;=\; \frac{\dim V}{16\pi^2}\,\ell_{\rm sp}^{-2}, \qquad g_{YM}^{-2} \;=\; \frac{1}{12\cdot 16\pi^2}\,\log\frac{\Lambda}{\mu}, \tag{8}$$

follows from $c_{EH} \propto \dim V\,\Lambda^2$ and $c_{YM} \propto \log(\Lambda/\mu)$ with $\Lambda \sim \ell_{\rm sp}^{-1}$: the ratio $8\pi G_N = c_{YM}/c_{EH} \propto \ell_{\rm sp}^2 \log(\Lambda/\mu)/\dim V$ grows with $\log\Lambda$ at fixed $\ell_{\rm sp}$, while $G_N^{-1} \sim \ell_{\rm sp}^{-2}$ is the dominant quadratic contribution. In particular $G_N \sim \ell_{\rm sp}^2$ (not $\ell_{\rm sp}^{-2}$). Equation (7) is Theorem 1 of [11] and holds unchanged since $\delta_A S_\Pi$ does not involve the a_2 sector. □

Remark 3.3. The sign in (6) reflects the Euclidean signature convention of the spectral derivation: the Yang–Mills energy density is positive, and the Yang–Mills stress tensor acts as a positive source of curvature, as expected. The coupling constant $8\pi G_N$ is *not an input* but is determined by the ratio of Seeley–DeWitt coefficients, both of which are fixed by the same spectral cutoff $\ell_{\rm sp}$.

4 Gauge–gravity coupling at order a_6

4.1 The a_6 Seeley–DeWitt coefficient

The a_6 coefficient of the operator $\mathcal{A}_{g,A}$ is a local integral of terms of total derivative order six. For a Laplace-type operator on a vector bundle with curvature endomorphism E and connection curvature Ω_A, the general form is [13, 12]:

$$\begin{aligned} a_6[\mathcal{A}_{g,A}] = \frac{1}{16\pi^2}\int_M \Big[&\beta_1 R^3 + \beta_2 R\,R_{\mu\nu}R^{\mu\nu} + \beta_3 R_{\mu\nu}{}^{\rho\sigma}R_{\rho\sigma}{}^{\alpha\beta}R_{\alpha\beta}{}^{\mu\nu} \\ &+ \gamma_1\,R_{\mu\nu\rho\sigma}\,\mathrm{Tr}(\Omega_A^{\mu\nu}\,\Omega_A^{\rho\sigma}) + \gamma_2\,R\,\mathrm{Tr}(\Omega_A^2) + \gamma_3\,\mathrm{Tr}(\Omega_A^4) + \gamma_4\,\mathrm{Tr}((\nabla\Omega_A)^2)\Big]\sqrt{g}\,d^4x, \end{aligned} \tag{9}$$

where β_i and γ_i are numerical coefficients computable from the Seeley–DeWitt algorithm [13].

4.2 The leading cross-coupling term

The dynamically significant cross-coupling between gravity and gauge at order a_6 is encoded in the γ_1 term:

$$\mathcal{L}^{(6)}_{\rm cross} = \frac{\gamma_1}{16\pi^2}\,R_{\mu\nu\rho\sigma}\,\mathrm{Tr}(F^{\mu\nu}\,F^{\rho\sigma})\,\sqrt{g}. \tag{10}$$

This term couples the Riemann tensor directly to the gauge curvature two-form and produces corrections to both the Einstein equation and the Yang–Mills equation:

Theorem 4.1 (a_6 cross-coupling)**.** *At order a_6, the joint equations of motion receive the corrections:*

$$G_{\mu\nu} = 8\pi G_N\,T^{YM}_{\mu\nu} \;+\; c_{(6)}\log\frac{\Lambda}{\mu}\Big[\nabla^\rho\nabla^\sigma\mathrm{Tr}(F_{\mu\rho}F_{\nu\sigma}) + \textit{(curvature terms)}\Big] + O(a_8), \tag{11}$$

$$D_\mu F^{a\mu\nu} = c'_{(6)}\log\frac{\Lambda}{\mu}\,[R^{\mu\nu\rho\sigma}, F_{\rho\sigma}]^a + O(F^3,\, R^2F), \tag{12}$$

where $c_{(6)}, c'_{(6)} > 0$ are scheme-dependent positive constants of order $\ell_{\rm sp}^2/16\pi^2$.

Proof. The corrections follow from the variation of the a_6 term (9) with respect to $g^{\mu\nu}$ and A^a_ν respectively. The metric variation of the γ_1 term uses the identity

$$\frac{\delta}{\delta g^{\mu\nu}} \int R_{\alpha\beta\rho\sigma}\,\mathrm{Tr}(F^{\alpha\beta}F^{\rho\sigma})\,\sqrt{g}\,d^4x \ni 2\,\nabla^\alpha\nabla^\beta\mathrm{Tr}(F_{\mu\alpha}F_{\nu\beta}) + (\text{curvature contractions}),$$

obtained by integration by parts of the Palatini variation. The connection variation of the γ_1 term gives the commutator structure in (12) through $\delta\Omega^{\mu\nu}_A = D^\mu(\delta A^\nu) - D^\nu(\delta A^\mu)$. The constants $c_{(6)}, c'_{(6)}$ are suppressed by two additional powers of $\ell_{\rm sp}$ relative to c_{EH} and c_{YM}, confirming the Seeley–DeWitt order counting. □

Remark 4.2. The corrections in Theorem 4.1 are suppressed by $\ell^2_{\rm sp}$ relative to the leading-order terms and are therefore negligible at scales $\ell \gg \ell_{\rm sp}$. At scales approaching $\ell_{\rm sp}$, the a_6 terms become comparable to the Born–Infeld saturation corrections and the EBI completion of Section 5 provides the appropriate non-linear resummation.

5 Eddington–Born–Infeld completion

5.1 Motivation and strategy

The induced gravity functional $c_{EH}a_2[g]$ diverges quadratically as $\Lambda \to \infty$, and the induced gauge functional $c_{YM}a_4[g, A]$ diverges logarithmically. In both sectors, the Born–Infeld saturation bound c_χ of the admissibility structure [3] provides a natural non-linear completion that regulates these divergences at the spectral length $\ell_{\rm sp}$. The gravitational EBI completion was established in [3, 7]; the present section constructs the gauge-sector completion and then assembles the joint EBI functional.

5.2 Gauge-sector Born–Infeld completion

In the gauge sector, the Born–Infeld saturation condition requires that the effective amplitude of the gauge field satisfies $A_n \le c_\chi/\sqrt{\lambda_n}$ at each spectral mode [3]. At the level of the field theory, the natural BI extension involves the combination $\delta^\mu_\nu + \ell^2_{\rm sp}F^\mu{}_\nu$; however, for a non-abelian $F^a_{\mu\nu}$, the antisymmetry $F_{\mu\nu} = -F_{\nu\mu}$ implies that $\sqrt{\det(\delta + \ell^2_{\rm sp}F)}$ is not automatically real (it equals the Pfaffian of F, which can change sign in Euclidean signature). The unambiguous non-abelian replacement uses the symmetric positive-semidefinite combination $M_{\mu\nu} \equiv F_{\mu\rho}F^\rho{}_\nu$, which is gauge-covariant and has non-negative eigenvalues. The gauge BI Lagrangian is therefore written as:

$$\mathcal{L}_{BI}[A] = \frac{1}{g^2_{BI}\,\ell^4_{\rm sp}}\left[\sqrt{\det\big(\delta^\mu_\nu + \ell^2_{\rm sp}\,M^\mu{}_\nu\big)} - 1\right], \qquad M^\mu{}_\nu = F^{a\mu\rho}F^a_{\rho\nu}, \tag{13}$$

which is manifestly symmetric in its matrix argument and reduces to the standard BI form in the abelian case.

Lemma 5.1 (Gauge BI reduces to Yang–Mills)**.** *In the limit $\ell^2_{\rm sp}M^\mu{}_\nu \to 0$:*

$$\mathcal{L}_{BI}[A] \;=\; -\tfrac{1}{4}\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu}) \;+\; O\big(\ell^4_{\rm sp}\mathrm{Tr}(M)^2\big)\,.$$

Proof. Expand $\sqrt{\det(I + \ell^2_{\rm sp}M)} = 1 + \frac{1}{2}\ell^2_{\rm sp}\mathrm{tr}(M) + O(\ell^4_{\rm sp})$. Since $\mathrm{tr}(M^\mu{}_\nu) = F^{a\mu\rho}F^a_{\rho\mu} = -\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$, the leading term is $-\frac{1}{2}\ell^4_{\rm sp}\,g^{-2}_{BI}\,\mathrm{Tr}(F^2)$, matching the Yang–Mills Lagrangian $-\frac{1}{4}\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$ with $g_{YM} = g_{BI}$. The symmetric argument M avoids the sign ambiguity of $\sqrt{\det(\delta + \ell^2 F)}$ in the non-abelian case. □

5.3 Joint Eddington–Born–Infeld functional

Definition 5.2 (Joint EBI functional)**.** *The joint Eddington–Born–Infeld extension of the projective spectral entropy functional is:*

$$S_{EBI}[g, A] = \frac{2}{\kappa^2} \int_M \left[\sqrt{\det\left(g_{\mu\nu} + \ell_{\mathrm{sp}}^2 \left(R_{\mu\nu} + \kappa F^a_{\mu\rho} F^{a\rho}{}_\nu\right)\right)} - \sqrt{g} \right] d^4x, \tag{14}$$

where $\kappa^2 = 8\pi G_N$ and the Ricci tensor $R_{\mu\nu}$ is computed from g.

Theorem 5.3 (EBI recovers joint Einstein–Yang–Mills at leading order)**.** *In the limit $\ell_{\mathrm{sp}}^2(R_{\mu\nu} + \kappa F^2_{\mu\nu}) \to 0$, the variation of $S_{EBI}[g, A]$ reproduces the joint system (6)–(7) at leading order. The EBI functional is regular at ℓ_{sp} in both sectors and satisfies the following boundary conditions:*

(EBI-1) $S_{EBI}[g, A] \to c_{EH} a_2[g] + c_{YM} a_4[g, A]$ *as* $\ell_{\mathrm{sp}} \to 0$*;*

(EBI-2) $S_{EBI}[g, A]$ *is bounded above by* $\ell_{\mathrm{sp}}^{-4}\mathrm{Vol}(M)$*;*

(EBI-3) S_{EBI} *is a minimal admissible diffeomorphism-invariant completion satisfying (EBI-1) and (EBI-2) within the tensorial class in which it depends on g and A only through the combination $g_{\mu\nu} + \ell_{\mathrm{sp}}^2(R_{\mu\nu} + \kappa F^2_{\mu\nu})$; uniqueness within this class is partial, subject to the residual ambiguity in the precise form of $\Xi_{\mu\nu}$ noted in [3].*

Proof. Expanding the determinant in (14):

$$\sqrt{\det(g + \ell_{\mathrm{sp}}^2 \Xi)} = \sqrt{g}\left[1 + \tfrac{1}{2}\ell_{\mathrm{sp}}^2 g^{\mu\nu}\Xi_{\mu\nu} - \tfrac{1}{4}\ell_{\mathrm{sp}}^4\left(\tfrac{1}{2}(g^{\mu\nu}\Xi_{\mu\nu})^2 - g^{\mu\rho}g^{\nu\sigma}\Xi_{\mu\nu}\Xi_{\rho\sigma}\right) + O(\ell_{\mathrm{sp}}^6\Xi^3)\right],$$

where $\Xi_{\mu\nu} = R_{\mu\nu} + \kappa F^a_{\mu\rho} F^{a\rho}{}_\nu$. The ℓ_{sp}^2 term gives $\kappa^{-2}\ell_{\mathrm{sp}}^2 R\sqrt{g}/2$ for the Ricci scalar part, matching $c_{EH}a_2[g]$ under the identification $\ell_{\mathrm{sp}}^2 = \kappa^2 c_{EH}/2$. The gauge contribution at this order is $\ell_{\mathrm{sp}}^2 F^2\sqrt{g}$, which matches $c_{YM}a_4$ upon renormalization. Property (EBI-2) follows from the bound $|\det(I + X)| \le e^{\mathrm{tr}(X)}$ applied to $X = \ell_{\mathrm{sp}}^2 g^{-1}\Xi$. Uniqueness within the ansatz class is partial; property (EBI-3) follows by the argument of [3] applied to the joint tensor $\Xi_{\mu\nu}$, up to the residual ambiguity in the traceless modification of $R_{\mu\nu}$. □

Remark 5.4. The joint EBI functional (14) is the Cosmochrony analogue of the Bañados–Ferreira–Skordis theory [1] and of the Eddington action, extended to include gauge curvature. The admissibility saturation bound c_χ provides the physical justification for the BI structure: the effective amplitude of any admissible mode is bounded above by $c_\chi/\sqrt{\lambda_n}$, which at the field-theory level becomes the saturation of the determinant form. The residual ambiguity in the precise form of $\Xi_{\mu\nu}$ (whether to use $R_{\mu\nu}$ or $R_{\mu\nu} - \frac{1}{4}g_{\mu\nu}R$) noted in [3] persists in the joint functional and is the primary open point of the completion.

6 The gauge–gravity hierarchy ratio

6.1 The two induced couplings

From [7] and [11], the two induced couplings of the joint functional are given in (8). Both emerge from the same spectral cutoff ℓ_{sp} and the same renormalization group flow; they differ in the UV divergence structure of their respective Seeley–DeWitt coefficients (Λ^2 versus $\log\Lambda$). In particular:

$$G_N \sim \frac{16\pi^2}{\dim V}\,\ell_{\mathrm{sp}}^2 \qquad \text{and} \qquad g_{YM}^2 \sim \frac{12 \cdot 16\pi^2}{\log(\Lambda/\mu)}. \tag{15}$$

6.2 The hierarchy ratio

Proposition 6.1 (Gauge–gravity hierarchy)**.** *The ratio of the two induced couplings is:*

$$G_N\, g_{YM}^2 \;=\; \frac{12\,\ell_{\rm sp}^2}{\dim V} \cdot \left[\log \frac{\Lambda}{\mu}\right]^{-1}. \tag{16}$$

For $\ell_{\rm sp} < \ell_{\rm Planck}$ and $\Lambda/\mu \gg 1$:

$$G_N\, g_{YM}^2 \;\ll\; 1. \tag{17}$$

Proof. Substitute (15) and multiply:

$$G_N\, g_{YM}^2 \;\sim\; \frac{16\pi^2\,\ell_{\rm sp}^2}{\dim V} \cdot \frac{12 \cdot 16\pi^2}{(16\pi^2)^2\, \log(\Lambda/\mu)} \;=\; \frac{12\,\ell_{\rm sp}^2}{\dim V\, \log(\Lambda/\mu)}.$$

The numerical factors of $16\pi^2$ cancel between G_N and g_{YM}^2. With $\ell_{\rm sp} \sim 10^{-35}$ m and $\log(\Lambda/\mu) \sim \mathcal{O}(10^2)$ at electroweak scale, and $\dim V \geq 4$ for $G_\Pi \supseteq \mathrm{SU}(2)\times\mathrm{U}(1)$, the ratio is suppressed by $\ell_{\rm sp}^2 \sim 10^{-70}$ m^2, confirming (17). □

6.3 Structural interpretation

The gauge–gravity hierarchy (17) is a structural consequence of the Seeley–DeWitt order separation between the two sectors:

$$\underbrace{a_2 \;\longrightarrow\; \text{gravity},\; G_N^{-1} \sim \ell_{\rm sp}^{-2}}_{\text{quadratic UV divergence, non-renormalizable}} \qquad \underbrace{a_4 \;\longrightarrow\; \text{gauge},\; g_{YM}^{-2} \sim \log\Lambda}_{\text{logarithmic UV divergence, renormalizable}}\;. \tag{18}$$

The single spectral cutoff $\ell_{\rm sp}$ generates two qualitatively different divergence structures. The hierarchy $G_N g_{YM}^2 \ll 1$ is therefore a consequence of the spectral stratification of the theory and requires no separate fine-tuning or anthropic selection.

Remark 6.2. The $\dim V$ factor in (16) provides a mild sensitivity of the hierarchy ratio to the gauge group. For $G_\Pi = \mathrm{SU}(2)\times\mathrm{U}(1)$ (unconditional [6]), $\dim V = 4$; for $G_\Pi = \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ (conditional on [H-color] [10, 9]), $\dim V = 12$ in the fundamental representation. The ratio (16) is thus three times smaller for the full Standard Model gauge group, increasing the hierarchy suppression.

7 Epistemic status and conditionalities

7.1 Status of each result

Table 1 collects the epistemic status of the main results of this paper.

7.2 Inherited conditionality on [H-color]

The identification $G_\Pi = \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ is taken as input from the spectral admissibility sub-programme. The SU(3) sector is established unconditionally for the colour-adapted Cayley graph in [10], and conditionally on hypothesis [H-color] (triplet co-admissibility) for the standard graph [10, 9]. All results of the present paper are independent of [H-color]; the conditionality affects only the value of $\dim V$ used in the hierarchy ratio.

Result	Status	Dependence
$\delta_g S_\Pi \ni c_{YM} T^{YM}_{\mu\nu}$ (Lemma 3.1)	Theorem-level	Metric-independence of $F_{\mu\nu}$
Coupled EYM system (Theorem 3.2)	Theorem-level	Lemma 3.1, [7, 11]
a_6 cross-coupling (Theorem 4.1)	Standard + structural	[13]
EBI completion (Theorem 5.3)	Structural, uniqueness ansatz	[3]
$\Xi_{\mu\nu}$ ambiguity in EBI	Open	Tensorial completion
Hierarchy ratio (Proposition 6.1)	Structural prediction	[7, 11]
$G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$	Conditional on [H-color]	[10, 9]

Table 1: Epistemic status of the main results.

7.3 Open points

The following open points are carried forward:

1. **EBI ambiguity.** The precise form of the tensor $\Xi_{\mu\nu}$ in the joint EBI functional (whether to use $R_{\mu\nu}$, $R_{\mu\nu} - \frac{1}{4} g_{\mu\nu} R$, or another traceless modification) is not uniquely fixed at the level of the present paper. This ambiguity is inherited from the gravitational BI completion of [3] and affects only subleading terms in the IR expansion.

2. **Quantitative a_6 coefficients.** The numerical values of $\gamma_1, \gamma_2, \ldots$ in (9) depend on the specific operator $\mathcal{A}_{g,A}$ and require explicit computation for the admissible bundle. This is deferred to a companion numerical study.

3. **Lorentzian continuation.** All spectral derivations are performed in Euclidean signature. The Lorentzian continuation of the joint EBI functional is established at the level of Q5b [4] for the geometric sector; its extension to the gauge sector remains an open step.

4. SU(3) **Yang–Mills uniqueness.** The analogue of Theorem 3.2 for the internal structure of the SU(3) sector (colour charge structure, structure constants) requires the proof of [H-color] and is identified as an open problem in [10].

8 Conclusion

The main result of this paper is the derivation of the coupled Einstein–Yang–Mills system (6)–(7) as the joint variational problem of the projective spectral entropy functional $S_\Pi[g, A]$, completing the synthesis announced in [11].

The central structural observation. The joint functional $S_\Pi[g, A] = \frac{1}{2} \log \det' \mathcal{A}_{g,A}$ encodes both gravitational and gauge dynamics in its Seeley–DeWitt expansion:

$$\begin{aligned} a_2 &\longrightarrow \text{Einstein gravity}, \quad G_{\mu\nu} = 0 \text{ (in vacuum)}, \\ a_4 &\longrightarrow \text{Yang–Mills gauge dynamics}, \quad D_\mu F^{a\mu\nu} = 0, \\ a_4 \text{ under } \delta_g &\longrightarrow \text{gauge–matter coupling to gravity}, \quad G_{\mu\nu} = 8\pi G_N T^{YM}_{\mu\nu}, \\ a_6 &\longrightarrow \text{gauge–gravity cross-coupling}, \quad R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}. \end{aligned}$$

The hierarchy of interactions is identified with a hierarchy of spectral order. This identification is the conceptually durable contribution of the present trilogy: gravity and gauge dynamics are not unified by enlarging the symmetry group or adding fields, but are *stratified* by the Seeley–DeWitt level at which the same operator $\mathcal{A}_{g,A}$ responds to variation. The different UV behaviours of the two sectors — quadratic divergence for gravity, logarithmic divergence for gauge — are a direct consequence of this spectral stratification, as is the gauge–gravity hierarchy $G_N g_{YM}^2 \ll 1$, which is structurally suppressed in $\ell_{\rm sp}^2$ and $[\log(\Lambda/\mu)]^{-1}$ without fine-tuning.

What the programme is. This trilogy (Gravity 3.0 [7], Q12 [11], Q13) can now be stated in one sentence: *the Cosmochrony programme is a spectrally stratified theory of emergent interactions, in which the type of a fundamental interaction is determined by the Seeley–DeWitt order at which the admissible operator responds.* The conventional question "what symmetry unifies gravity and gauge?" is replaced by: "at what spectral order does the projection respond?"

Remaining open points. The Eddington–Born–Infeld completion of Section 5 provides a unified non-linear regulator at $\ell_{\rm sp}$, recovering both sector-specific completions as limits. The a_6 cross-coupling of Section 4 is suppressed by $\ell_{\rm sp}^2$ and becomes relevant only near $\ell_{\rm sp}$. The residual open points — EBI tensor ambiguity, a_6 numerical coefficients, Lorentzian continuation, SU(3) Yang–Mills uniqueness — are each well-posed and bounded in scope. They do not affect the structural hierarchy result (16) or the coupled equations of motion (6)–(7). The natural continuation of the programme is the treatment of fermionic matter and chirality, where the spectral stratification principle predicts the existence of a dedicated spectral level carrying spinor structure.

References

[1] M. Bañados and P. G. Ferreira. Eddington's theory of gravity and its progeny. *Phys. Rev. Lett.*, 105:011101, 2010.

[2] J. Beau. Admissible non-injective transitions as the primitive of physical description, 2026. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.19616329.

[3] J. Beau. Bounded relaxation and the dynamical selection of spacetime geometry, 2026. Cosmochrony companion paper C about Born-Infeld, doi:10.5281/zenodo.18407505.

[4] J. Beau. Effective spacetime geometry from admissible non-injective projection, 2026. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.19744005.

[5] J. Beau. From admissibility to quantum structure: Phase coherence and correlations from non-injective projection, 2026. spectral admissibility subprogram "Q1" paper, doi:10.5281/zenodo.19561060.

[6] J. Beau. Gauge structure from admissible non-injective projection, 2026. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19655294.

[7] J. Beau. Infrared einstein response from a renormalized spectral entropy functional, 2026. Cosmochrony companion paper H, doi:10.5281/zenodo.18818721.

[8] J. Beau. Quantum structure beyond spin-1/2: Co-admissible sectors and the emergence of su(2) as a fixed point authors/creators, 2026. spectral admissibility subprogram "Q2" paper, doi:10.5281/zenodo.19616444.

[9] Jérôme Beau. Colour triplet co-admissibility on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: Numerical test of hypothesis [H-color], 2026. "O32" paper, doi:10.5281/zenodo.20145607.

[10] Jérôme Beau. Su(3) from colour triplet co-admissibility: Towards the complete standard model gauge group, 2026. "O31" paper, doi:10.5281/zenodo.20030491.

[11] Jérôme Beau. Yang–mills dynamics from projective spectral entropy: Vertical heat-kernel variation on the admissible fibre, 2026. "Q12" paper, doi:10.5281/zenodo.20189137.

[12] L. E. Parker and D. J. Toms. *Quantum Field Theory in Curved Spacetime: Quantized Fields and Gravity*. Cambridge University Press, 2009.

[13] D. V. Vassilevich. Heat kernel expansion: user's manual. *Phys. Rep.*, 388:279–360, 2003.